



Prosthetic Fingers for Those With Digit Amputations

Elaine Rama, Sierra Manriquez, Erik Prestin, Ben Delozier

ENG 1102: L34 Team 5



Introduction

Fingers play a key role in the functionality of the hand. Prosthetic fingers offer those who are missing fingers a chance to regain independence and functionality. This improves the quality of life of the user by restoring essential everyday tasks such as gripping, writing, and driving. The team designed an assembly of manipulatable prosthetic fingers to help users lead more fulfilling lives despite the challenges faced due to finger loss.

The planned use of the prosthetic is to assist those with thumb, index, and ring finger amputations. The device is designed to provide both comfort and precision, using sensors and motors to enable the prosthetic fingers to move on the user's command. This allows the prosthetic to move more fluidly, aligning with the user's natural movement.

Societal and Environmental Impacts

Social Impacts:

❑ The prosthetic is designed to be discreet, disguised in an aesthetically pleasing glove that helps users fit in socially. The glove effectively conceals the prosthetic, allowing the users to perform everyday tasks without drawing unwanted attention. The team chose the ring finger for its societal significance, while the index and thumb were chosen for being the most amputated fingers.

Environmental Impacts:

❑ The prosthetic is made from 3D-printed plastic, which contributes to the growing issue of plastic waste and pollution.

Target Users

Target Users include:

❑ Digit Amputees without thumb, index, and ring finger on the left hand.

Design Context and Considerations

The team's objective was to create a working prototype to be tested under real-world conditions. Our design needed to:

- ❑ Prioritize user needs, comfort, and preferences
- ❑ Enhance quality of life
- ❑ Allow inclusivity of amputees in society
- ❑ Enhance physical and psychological well-being of user
- ❑ Empower User
- ❑ Closely replicate natural finger movements

Design Considerations:

- ❑ The team's Failure Modes and Effects Analysis (FMEA) identified failure modes, symptoms, effects, severity, and frequency for each component of the prosthetic fingers. This provided new constraints and considerations that the prosthetic needed to follow. After making necessary design changes, the team decreased possible risks by 56.4%.
- ❑ W.H.O prosthetic standards: Durable, functional, easy to put on and remove, safe and working technologies, products easily adjusted and maintained.
- ❑ Research conducted showed thumb and index fingers as most amputated. While the ring finger was chosen due to its societal and emotional significance.

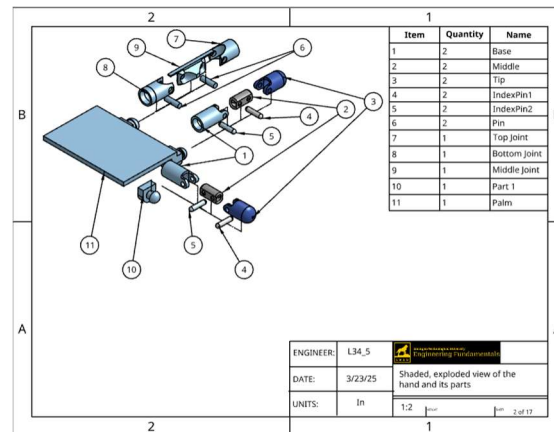


Fig. 2 Final Model Assembly

Detailed Views

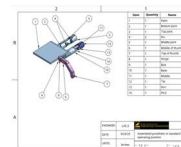


Fig. 4 Initial Model Assembly

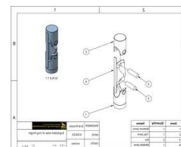


Fig. 5 Ring Finger Final Model

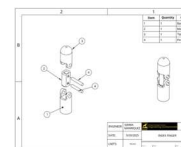
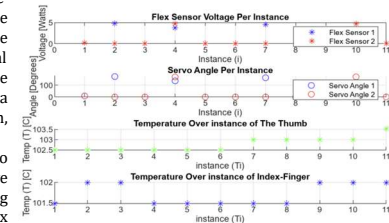


Fig. 6 Index Finger and Thumb Final Model

Results and Discussion

The team's control program is designed to intake data from the flex sensors and thermistors, then analyze it to operate the corresponding servo motors and buzzer outputs. The thermistors collect temperature data from their corresponding servo motors as a safety feature to ensure that the servo motors do not overheat and harm users. If a servo motor's temperature reaches a heat of 35 degrees Celsius or higher, the buzzer will sound, alerting the user to immediately turn off the prosthetic. Two graphs show the temperatures of the motors in real time as a visualization (Fig.1). The flex sensors collect voltage data from the user's movement. Then, the voltage is proportionally converted to an angle. The servo motor uses this angle to move according to the corresponding voltage. Two graphs show flex sensor voltage and servo angle in real-time as a visualization (Fig.1).



Prosthetic Assembly

List of components:

- ❑ Passive Buzzer
- ❑ 830 Tie-Points Breadboard
- ❑ Flex Sensor
- ❑ UNO R3 Controller Board
- ❑ Servo Motor
- ❑ USB Cable
- ❑ String
- ❑ Thermistor
- ❑ 3D Print
- ❑ Glove

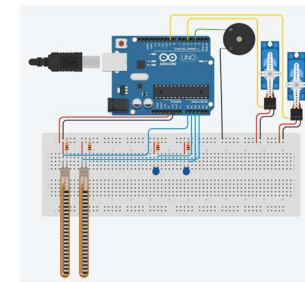


Fig. 3 Arduino Wiring Diagram

Conclusion

The team designed prosthetic fingers to restore the everyday comforts that amputees deserve. The team carefully considered the societal and environmental impacts to enhance the design. Through extensive testing and multiple finger design iterations, the team refined the prosthetic, creating a final design that balances functionality, comfort, and user preferences.